

Vibe Shuffle: Browser-Based Affective Music Selection from Facial Valence and Cardiac Arousal

Markus Baumann

Ludwig-Maximilians-Universität München

Munich, Germany

markus.baumann@campus.lmu.de

Abstract

Music is widely used to regulate emotion, yet mainstream recommenders optimise long-run engagement rather than a listener’s momentary state. We present *Vibe Shuffle*, a browser-based research application that estimates a baseline-relative valence–arousal coordinate: facial blendshapes and custom rules supply Valence, while optional heart rate and short-term heart-rate variability supply Arousal. The adaptive policy selects the nearest unplayed track in the matching quadrant of a fixed 100-track catalog. A participant-blinded, counterbalanced crossover compares this policy with seeded Random selection from the same pool. Each track yields 7-point mood-fit and liking ratings plus a categorical mood report. Raw camera frames and heart-rate packets remain in browser memory; the participant-triggered CSV contains derived signals, ratings, pseudonymous identifiers, and timestamps. The open artifact makes the heuristics, quality rules, selection mechanism, and limitations auditable.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Information systems** → *Recommender systems*.

Keywords

affective computing, music recommendation, valence–arousal, facial expression, heart-rate variability, on-device processing

1 Introduction

People use music to maintain or change how they feel, and emotional fit affects listening satisfaction [2]. Most streaming recommenders, however, learn long-term taste rather than react to a listener’s current state. Affective computing offers facial and autonomic signals, but these measurements are noisy, privacy-sensitive, and easy to overinterpret.

Vibe Shuffle is a browser-based research prototype for testing a narrow claim: does a transparent state-adaptive song policy produce higher perceived mood fit than Random selection from the same controlled pool? The prototype represents both listener and song in Russell’s Valence–Arousal space [5, 6]. It does not claim to recover objective or clinical emotion.

The contribution is an auditable end-to-end artifact: client-side signal processing, a frozen 100-track catalog, an explicit nearest-track policy, a participant-blinded crossover protocol, and a row-level CSV audit trail. Raw camera frames and heart-rate notifications are not uploaded by the application; derived measurements and pseudonymous study metadata are exported only through a deliberate local download.

2 Methodology

2.1 System overview

Vibe Shuffle is a React/Vite single-page application without an application backend. A camera channel supplies horizontal Valence, an optional Bluetooth Heart Rate Service channel supplies vertical Arousal, and a 60 s listening window is summarized into one listener coordinate. The next song is selected from a fixed catalog and played with Spotify’s Web Playback SDK. The participant then supplies ratings before the protocol advances.

MediaPipe and Spotify remain external dependencies: model assets, authentication, and playback require network traffic. Signal inference and CSV construction occur in the browser.

2.2 Facial display and Valence

At a target interval of 120 ms, MediaPipe Face Landmarker [3] returns 52 blendshape activations. The application reduces smile, cheek-squint, frown, brow, eye, jaw, and mouth cues to custom *Happy*, *Sad*, and *Tense* display scores. MediaPipe does not output these emotion labels; the mapping is a hand-specified, FACS-inspired heuristic in the application source.

A 14 s calibration captures the participant’s current facial reference. A slow running baseline and exponential smoothing reduce resting-morphology and single-frame effects. If $\bar{H}$, $\bar{S}$, and $\bar{T}$ are the mean baseline-relative rule scores for a listening window, Valence is

$$V = \text{clip}_{[0.05, 0.95]} \left[0.5 + 1.05 (\bar{H} - 0.55\bar{S} - 0.45\bar{T}) \right]. \quad (1)$$

No detected face sets $V = 0.5$. Whole-frame movement and rhythmic head motion currently add positive evidence to the facial channel. This encodes an exploratory engagement assumption, not an established affect measurement, and is treated as a limitation.

2.3 Cardiac signal and Arousal

The optional sensor exposes BPM and, when available, RR intervals: elapsed time between successive cardiac beats. The application rejects RR values outside 300–2000 ms and values that differ by more than 30% from the previous accepted interval. RMSSD measures short-term variability,



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$$\text{RMSSD} = \sqrt{\frac{1}{n-1} \sum_{i=1}^{n-1} (RR_{i+1} - RR_i)^2}. \quad (2)$$

At least 20 accepted RR intervals are required for the full HR/RMSSD estimate. A 120 s personal baseline stores median HR and median chunk-level RMSSD. Robust spread is $\max(1.4826 \text{ MAD}, 4 \text{ BPM})$ for HR and $\max(1.4826 \text{ MAD}, 6 \text{ ms})$ for RMSSD. The baseline-relative scores are

$$z_{HR} = \frac{\text{median}(HR_w) - \text{median}(HR_b)}{s_{HR}}, \quad (3)$$

$$z_{RMSSD} = \frac{\text{median}(RMSSD_b) - RMSSD_w}{s_{RMSSD}}. \quad (4)$$

Positive z_{HR} and z_{RMSSD} therefore both represent higher-Arousal evidence. A deadband discards $|z| \leq 0.25$ and caps larger magnitudes at three spread units. With this transform D , the implemented physiological coordinate is

$$A_{phys} = \text{clip}_{[0,1]} [0.5 + 0.18 (0.75D(z_{HR}) + 0.35D(z_{RMSSD}))]. \quad (5)$$

When RMSSD suggests higher Arousal but HR is at or below baseline, the RMSSD weight is multiplied by 0.2. This conservative conflict rule limits unstable short-window HRV. SDNN and a rhythm-coherence value are exported as diagnostics but do not drive Arousal. The application receives processed RR data, not a raw ECG waveform, and cannot identify every motion artifact.

2.4 Fusion and windowing

Facial display supplies V . Usable physiology supplies the Arousal base; positive camera movement can add upward Arousal. Without usable physiology, camera motion supplies upward-only Arousal. Without a face, $V = 0.5$; without either usable signal, the coordinate is $(0.5, 0.5)$.

During confirmed playback, every change to the live fused state appends a point to the trial buffer. At rating time, the application takes the arithmetic mean of all valid buffered positions. The sampling is event-driven rather than fixed-rate and the mean is not time-weighted. The export therefore includes the sample count and actual listening duration so this approximation can be audited.

Splitting both axes at 0.5 yields four recommendation quadrants: Energetic (high V , high A), Calm (high V , low A), Tense (low V , high A), and Melancholic (low V , low A).

2.5 Study catalog

The frozen catalog contains 100 real Spotify track IDs and historical Valence, Energy, Instrumentalness, and popularity values from the public Kaggle Spotify Tracks Dataset [4]. High-popularity entries were de-duplicated, manually curated, and frozen at 25 tracks per quadrant. The final artifact is enforced by a catalog integrity test; the repository does not claim a fully automatic generator for the curatorial choices.

Song Valence supplies the horizontal coordinate and song Energy is treated as the catalog-side counterpart of listener Arousal. This is a modeling assumption: musical Energy and physiological Arousal

Table 1: Fixed catalog classification.

Label	Rule	Tracks
Energetic	$V_s \geq 0.5, E_s \geq 0.5$	25
Calm	$V_s \geq 0.5, E_s < 0.5$	25
Tense	$V_s < 0.5, E_s \geq 0.5$	25
Melancholic	$V_s < 0.5, E_s < 0.5$	25

are related constructs, not identical measurements. The current catalog is mainstream rather than instrumental-only.

2.6 Track selection

Both conditions hard-exclude the current and previously played tracks. Random ranks every remaining track with a deterministic pseudorandom score from track ID and session seed; measured signals are logged but ignored.

Vibe maps the listener target (V, A) to its quadrant, keeps unplayed tracks in that quadrant, and minimizes Euclidean distance,

$$s^* = \arg \min_{s \in Q(V,A)} \sqrt{(V_s - V)^2 + (E_s - A)^2}. \quad (6)$$

The seed resolves only an exact distance tie. If the quadrant is exhausted, the method falls back to all remaining tracks. The first track uses the setup-time coordinate; subsequent Vibe selections use the mean coordinate from the preceding trial. The CSV records target, source, distance, and quadrant match.

2.7 Validation protocol

Every participant completes both conditions in a participant-blinded crossover: one five-track Random block and one five-track Vibe block. Protocol 1 uses Random-Vibe; Protocol 2 uses Vibe-Random. Odd/even participant numbers suggest alternating orders, which the experimenter can override before start. This is within-participant comparison with order counterbalanced across participants, not double blinding.

Each track plays for up to 60 s. An early *Rate now* action is preserved in the export. The mandatory prompt collects 7-point liking, 7-point mood fit, and a categorical current-mood report. Mood fit is the primary outcome; liking is a control for general song preference.

The proposed participant-level analysis averages the five mood-fit ratings per condition and evaluates the paired Vibe-minus-Random differences. With a small sample and ordinal responses, an exact paired Wilcoxon or paired permutation test is the robust primary option; a paired t -test is a sensitivity analysis when the difference distribution is sufficiently regular. Early and low-quality trials should be handled by a rule fixed before inspecting the condition result.

3 Artifact Output

One local CSV row is created per rated track. It contains protocol and song metadata, both ratings, categorical self-report, listening duration, fused state, expression summaries, HR/RR/HRV summaries, selection audit fields, and quality flags. Non-finite numeric values become blank cells rather than textual NaN values.

The repository intentionally includes no participant dataset. This paper documents the implemented system and analysis plan; it does not report an inferential participant result.

4 Discussion

The artifact isolates the policy comparison by holding the song pool, playback path, rating questions, and exposure count constant. Its transparent rules make failures auditable and provide a concrete basis for an HCI study of implicit adaptation.

The limits are substantial. Facial movement is not a direct readout of felt emotion [1]; the custom weights were not learned from labeled affect; HR/HRV reflects many non-emotional influences; and the personal baseline captures relative change rather than absolute initial mood. Motion contributes to both Valence and Arousal and can confound the two channels. The event-driven window mean is not time-weighted. The fixed balanced catalog aids control but limits ecological and catalog-scale generalization. Spotify Premium, Development-mode allowlisting, browser permissions, and BLE support constrain deployment.

The privacy claim is also bounded. The application has no study-data upload endpoint and does not export raw frames or packets, but GitHub, MediaPipe, and Spotify still receive ordinary service traffic. Exported participant numbers and timestamps are pseudonymous, not anonymous, and require appropriate research- data handling.

Future work should validate the mapping against a separate labeled dataset, compare motion ablations, use a time-weighted fusion window, prespecify artifact handling, and evaluate the recommendation effect with a larger sample and a catalog not manually balanced by quadrant.

5 Conclusion

Vibe Shuffle is an open, browser-based testbed for comparing a transparent state-adaptive music policy with Random selection under controlled stimuli. It does not solve emotion recognition; it makes one rule-based recommendation hypothesis explicit, measurable, and falsifiable through participant-rated mood fit.

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